

PEORIA, IL, USA

WMO: 725320

Lat: 40.668N

Lon: 89.684W

Elev: 198

StdP: 98.97

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 14842

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.6	-15.7	-23.4	0.5	-17.7	-20.8	0.6	-15.1	11.1	-1.3	10.2	-2.1	3.2	300	0.504

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.2	33.4	24.7	32.1	24.0	30.7	23.2	26.3	31.4	25.3	30.3	24.4	29.0	4.6	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.8	20.3	29.7	23.8	19.1	28.5	22.9	18.1	27.5	82.7	31.3	78.4	30.3	74.7	29.2	29.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.0	8.7	7.8	DB	-22.7	35.8	3.7	1.6	-25.4	36.9	-27.6	37.8	-29.7	38.7	-32.4	39.9	(4)
(5)			WB	-23.1	27.8	3.6	0.9	-25.7	28.4	-27.8	28.9	-29.9	29.4	-32.5	30.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.4	-3.8	-1.3	5.0	11.5	17.3	22.4	24.4	23.5	19.7	12.7	5.5	-0.8	(6)	
	DBStd	11.10	6.90	6.54	6.25	5.22	4.68	3.54	3.06	2.85	4.31	5.23	5.53	6.10	(7)	
	HDD10.0	1498	430	319	179	42	3	0	0	0	1	31	156	337	(8)	
	HDD18.3	3149	687	549	414	211	78	8	1	2	35	187	386	592	(9)	
	CDD10.0	2011	1	3	25	88	229	373	446	418	291	116	20	2	(10)	
	CDD18.3	621	0	0	1	7	46	131	188	161	75	12	0	0	(11)	
	CDH23.3	5611	0	0	10	77	399	1191	1844	1343	648	97	1	0	(12)	
	CDH26.7	1957	0	0	0	12	104	421	718	462	222	18	0	0	(13)	
(14) Wind	WSAvg	3.5	4.0	4.0	4.2	4.4	3.7	3.2	2.7	2.5	2.8	3.4	3.8	3.7	(14)	
(15) Precipitation	PrecAvg	929	44	42	70	98	105	100	97	83	90	72	68	59	(15)	
	PrecMax	1406	206	135	190	208	265	297	258	219	332	202	194	160	(16)	
	PrecMin	563	6	12	17	18	18	15	8	6	1	1	6	8	(17)	
	PrecStd	207	34	27	40	46	57	64	54	53	69	44	41	33	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.0	18.3	24.8	28.3	31.6	34.1	36.2	34.3	33.7	29.0	22.0	17.0	(19)	
		MCWB	11.4	12.7	17.0	18.2	22.0	24.0	25.7	25.0	23.4	20.6	15.5	14.6	(20)	
	2%	DB	11.2	14.3	20.9	25.7	29.4	32.4	33.7	32.6	31.4	26.0	18.6	13.3	(21)	
		MCWB	9.0	10.4	14.5	17.0	20.7	23.8	25.5	24.6	22.5	18.5	14.0	11.0	(22)	
	5%	DB	7.9	10.8	17.8	23.0	27.5	30.9	32.1	30.9	29.3	23.6	16.3	10.2	(23)	
		MCWB	5.9	7.8	13.0	15.5	19.6	22.8	24.6	23.7	21.2	17.3	12.9	8.1	(24)	
	10%	DB	4.9	7.5	14.7	20.5	25.6	29.2	30.5	29.3	27.3	21.1	14.0	7.5	(25)	
		MCWB	3.2	5.2	10.7	14.0	18.6	22.0	23.7	22.7	20.2	16.3	10.8	5.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.0	14.6	17.9	20.3	23.7	26.2	27.9	27.1	25.0	22.1	16.9	15.1	(27)	
		MCDB	14.1	16.9	23.1	26.4	29.2	31.7	33.1	32.2	31.1	26.9	19.8	16.8	(28)	
	2%	WB	9.4	11.1	15.7	18.2	22.3	25.0	26.6	25.9	23.8	20.3	15.1	11.5	(29)	
		MCDB	10.5	13.9	19.8	23.1	27.5	30.6	31.7	30.8	29.2	23.8	17.4	12.8	(30)	
	5%	WB	5.9	7.9	13.6	16.6	21.0	24.0	25.7	24.8	22.7	18.6	13.1	8.2	(31)	
		MCDB	7.8	10.6	17.3	21.3	25.6	29.2	30.5	29.2	27.2	21.9	15.6	9.8	(32)	
	10%	WB	3.2	5.2	10.7	15.0	19.8	23.1	24.8	23.8	21.5	17.0	11.0	5.5	(33)	
		MCDB	4.8	7.6	14.3	20.0	24.1	27.8	29.3	27.8	25.6	20.7	13.7	7.5	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	8.2	8.9	10.1	11.1	10.9	10.4	10.2	10.1	11.4	10.7	9.2	7.7	(35)	
		MCDBR	11.1	12.8	13.3	14.3	12.7	11.9	11.3	11.5	13.0	13.3	12.1	10.6	(36)	
	5% WB	MCWBR	9.2	9.6	8.5	8.1	6.3	5.5	4.9	5.2	5.6	7.0	8.5	8.8	(37)	
		MCWBR	10.8	12.0	12.3	12.4	11.2	10.6	10.1	10.1	11.1	10.6	10.9	9.9	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.294	0.315	0.340	0.389	0.426	0.440	0.453	0.436	0.402	0.351	0.307	0.293	(40)	
		taud	2.439	2.374	2.400	2.265	2.222	2.229	2.179	2.221	2.288	2.434	2.503	2.454	(41)	
	Ebn at Noon	Ebn at Noon	870	898	911	883	855	840	825	828	833	845	849	839	(42)	
		Edh at Noon	79	97	106	131	140	139	145	135	118	91	74	72	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.81	2.73	3.71	4.75	5.49	6.22	6.24	5.48	4.66	3.13	2.08	1.52	(44)	
	RadStd	RadStd	0.20	0.27	0.27	0.39	0.42	0.47	0.34	0.35	0.36	0.36	0.17	0.20	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+119	+74
Regional (44 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+97	+62

Nomenclature: See separate page